

Semantic Feedback-Based RAG for Radiology Report Generation

Xing Jia, Yun Xiong*, Songwen Pei, Yumeng Zhang, Cairong Yan, and Zhijun Fang

Abstract: Radiology report generation aims to produce textual reports automatically based on input images, a critical process that aids in accurate diagnoses and lightens the workload of radiologists. Following recent advances in Large Language Models (LLMs), several Retrieval-Augmented Generation (RAG) based report generation models have been proposed. Despite the continuously improved performance, these report generation models often suffer from two main limitations, i.e., interference of irrelevant information, and lack of alignment between the input image and the resulting generated report. In this study, we propose the Semantic feedback based RAG Radiology report generation model, namely RAGSemRad. RAGSemRad comprises two key components: the fine-grained semantic retrieval module and the semantic assessment module. The fine-grained semantic retrieval module is designed to retrieve adequate and relevant prompt information, while ignoring irrelevant interference. This is achieved by clustering the data at the semantic level and leveraging the domain knowledge within a large pre-trained visual-language model, thus alleviating the issues of hallucination and databias. Further, the semantic assessment module enhances the performance of the upper bound by enhancing the alignment between the input image and the resulting generated report, utilizing supervision signals derived from paired image-label data. Experimental evaluations are conducted on two benchmarks, IU X-Ray and MIMIC-CXR, to assess the performance of RAGSemRad. The results demonstrate RAGSemRad exhibits competitive performance compared to the state-of-the-art methods, showcasing its potential to advance automatic radiology report generation.

Key words: radiology report generation; Retrieval-Augmented Generation (RAG); semantic assessment

1 Introduction

In the realm of medicine, radiological images are extensively utilized in clinical settings for purposes such as diagnosis and treatment planning^[1]. The field

of automatic radiology report generation, which involves generating textual reports from medical images, has garnered significant interest lately due to its potential to aid in diagnosis and lighten the

- Xing Jia is with School of Computer Science and Technology, Donghua University, Shanghai 200000, China, and also with Shanghai Key Laboratory of Data Science, Shanghai 200000, China. E-mail: xjia@dhu.edu.cn.
- Yun Xiong is with School of Computer Science, Fudan University, Shanghai 200000, China. E-mail: yunx@fudan.edu.cn.
- Songwen Pei is with Faculty of Intelligence Technology, Shanghai Institute of Technology, Shanghai 201418, China. E-mail: swpei@usst.edu.cn.
- Yumeng Zhang is with School of Management, Guangdong University of Science and Technology, Guangzhou 523083, China. E-mail: zhangyumeng@gdust.edu.cn.
- Cairong Yan and Zhijun Fang are with School of Computer Science and Technology, Donghua University, Shanghai 200000, China. E-mail: cryan@dhu.edu.cn; zjfang@dhu.edu.cn.

* To whom correspondence should be addressed.

Manuscript received: 2025-02-16; revised: 2025-03-21; accepted: 2025-04-01

workload of radiologists. As illustrated in Fig. 1, the objective is to automatically produce reports for given images.

In recent years, there has been a surge of interest in report generation methods, inspired by tasks such as image captioning^[2] and paragraph generation^[3]. A number of methods have been proposed for report generation, as evidenced by a wide range of studies^[1, 4–19]. These methods have significantly advanced the field of report generation in the medical domain. The prevailing methods for generating reports typically adhere to the encoder-decoder framework, originally tailored for neural machine translation^[20]. In this framework, Convolutional Neural Networks (CNNs) are often employed as encoders to extract intricate feature representations from input images. Subsequently, decoders, such as Recurrent Neural Networks (RNNs) or Transformers, are utilized to transform these feature representations into textual reports. The main difference between image captioning (or paragraph generation) and report generation is that report generation pays more attention to the fundamental concern of clinical accuracy, i.e., the alignment between the input image and the resulting generated report. As shown in Fig. 1, the clinical accuracy of the sentence “heart size in normal” is bad, since the ground-truth is “heart size in enlarged”. Moreover, in the context of report generation, it is essential to emphasize abnormalities or diseases that are critical for assisting radiologists in making

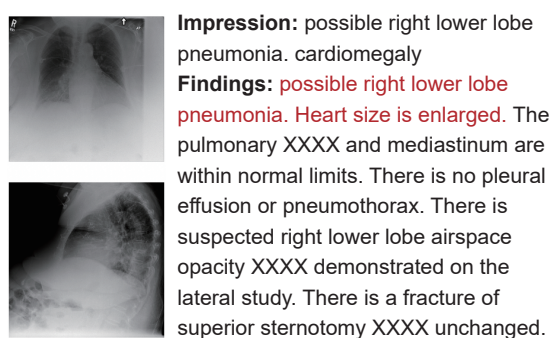


Fig. 1 An instance that involves a chest X-ray image paired with its corresponding report. Typically, a report comprises the “impression” section and the “findings” section. The “findings” segment delineates intricate details regarding both normal and abnormal observations. The “impression” segment offers a concise summary statement. For radiologists, a model focusing on descriptions of abnormalities or diseases are of greater concern to them, i.e., the abnormality enlarge heart-size.

informed clinical decisions. In other words, for a high-quality report, above all, it needs to be semantically correct. Furthermore, it should highlight abnormalities or diseases.

Existing methods in radiology report generation primarily focus on optimizing applying text-similarity objectives, such as teacher-forcing or reinforcement learning. However, text-similarity-driven strategies tend to result in generated reports with low clinical accuracy. A few studies have attempted to incorporate semantics in a implicit manner, e.g., Yan et al.^[21] introduced a weakly supervised contrastive loss based on clustering, and Jia et al.^[22] introduced a semantic weighted contrastive loss by setting different hard levels of negative samples. Meanwhile, the benchmarks used for report generation exhibit imbalance, where there is a significant prevalence of normal descriptions compared to abnormal ones. As a result, existing models tend to generate plausible yet routine reports, often failing to effectively highlight abnormalities. In order to address this issue, several studies have explored the utilization of knowledge graphs^[7, 10, 19, 23], and involves inferring abnormalities based on the correlations among disease nodes. Nonetheless, the limited number of disease nodes within the knowledge graph hinders its overall effectiveness and capability. Also, the process of constructing the knowledge graph is quite tedious.

Recently, following the advance of Large Language Models (LLMs), several Retrieval-Augmented Generation (RAG) based report generation models have been proposed^[24–27], e.g., ChatCAD^[26] integrates a multi-component classification network, a segmentation model, and a report generator with ChatGPT for report generation. EALBEF^[28] performs an effective retrieval step by integrating ALBEF^[24] with a semantic segmentation network, which allows the embeddings alignment of the ALBEF network to focus on the interesting portion of input images. Despite the continuously improved performance, these report generation models often suffer from two main limitations:

- **Interference of irrelevant information.** Current RAG-based report generation methods typically retrieve the entire report or only the summary content (i.e., the “impression” part shown in Fig. 1) during the retrieval phase, thereby being interfered by irrelevant or inadequate prompt information^[27, 28]. This often

leads to serious hallucination in the generated reports, affecting the overall quality of the output.

- **Lack of explicit semantic supervision.** As we have mentioned before, alignment between the input image and the resulting generated report is the fundamental concern. Therefore, having semantic guidance is crucial for achieving this goal, but current RAG-based report generation methods do not operate in this way^[25–28]. And they may struggle to capture the underlying semantics of generated reports, and further lead to a model generating text-similarity-oriented reports at the inference phase, without considering the alignment between the image and generated report.

To overcome the limitations, we propose the Semantic feedback based RAG Radiology report generation model, namely RAGSemRad. RAGSemRad consists of two main components:

- **Fine-grained semantic retrieval module.** Although sentences describing abnormalities are a smaller subset of benchmark datasets compared to normal descriptions, they encompass a wide array of medical diagnoses^[29]. Considering this, we introduce the fine-grained semantic retrieval module. By clustering the data at the semantic level and utilizing the domain knowledge within a large pre-trained visual-language model, our module achieves the purpose of retrieving adequate and relevant prompt information, while ignoring irrelevant interference, thereby alleviating the issues of hallucinations and data bias.

- **Semantic assessment module.** In clinical settings, each Chest X-Ray (CXR) image requires an accompanying semantic description to fulfill its diagnostic purpose. Beyond paired image-report data, datasets structured in the form of image-label pairs remain underexplored. Therefore, we introduce the semantic assessment module. The core concept behind this module is to enhance the performance upper bound by enhancing the alignment between the input image and the resulting output report, leveraging the supervision signals from additional paired image-label data.

In summary, the main contributions of this paper can be outlined as follows:

- A novel radiology report generation model, RAGSemRad, is proposed, advocating a pragmatic approach to enhance the performance of report generation.

- The fine-grained semantic retrieval module is introduced to alleviate the issues of hallucination and data bias through a more feasible approach, specifically by retrieving sentence-level semantic information within the dataset itself.

- The semantic assessment module is introduced, aimed at enhancing the upper bound of the model's performance by leveraging the additional image-label data, thereby reducing reliance on the paired image-report data.

- Experiments are conducted on two real-world datasets, with evaluation based on natural language generation and clinical efficacy metrics. Competitive results are demonstrated by RAGSemRad when compared to the state-of-the-art methods.

2 Related Work

2.1 Image caption

Image captioning, which involves understanding a given image by generating a descriptive sentence, has been extensively researched in recent years^[2, 30–35]. Progress in image captioning has outpaced that in report generation due to advancements in deep learning techniques and the availability of large-scale datasets. Traditional image captioning methods typically adopt an encoder-decoder architecture, where CNNs encode the image to derive corresponding representations, and RNNs coupled with attention mechanisms translate these representations into captions^[2, 31, 33, 35]. Additionally, some studies have explored visual relationships among image regions by constructing scene graphs during the encoding phase^[30, 34]. Given the limitations of RNNs in modeling long-range dependencies, Transformer-based models have been introduced in image captioning tasks^[32, 34]. In contrast to image captioning, paragraph generation, a closely related task to report generation, focuses on producing detailed and semantically coherent paragraphs describing visual content. Hierarchical RNN structures are commonly employed in paragraph generation, where the decoding stage is divided into two steps: topic vector generation by a topic decoder and sentence generation by a sentence decoder. Concurrently, the advent of LLMs has spurred attempts by researchers to harness these models for the generation of textual paragraphs^[36, 37], leveraging the contextual comprehension inherent in LLMs. Illustratively, Xie et al.^[37] employed a substantial pre-trained multi-modal

model to derive visual cues from input images, which are then used to craft detailed paragraphs.

2.2 Radiology report generation

Compared to image captioning and paragraph generation tasks, ensuring clinical accuracy in generated reports is the fundamental concern in the radiology report generation task. Most report generation models adhere to the encoder-decoder framework^[1, 4–19, 38, 39], addressing differences between image captioning (or paragraph generation) and report generation. For example, Jing et al.^[11] introduced a co-attention mechanism into report generation models to simultaneously explore visual and semantic information. Wang et al.^[40] proposed a self-boosting framework that enhances report generation through collaboration between the primary task of report generation and an auxiliary task of image-text matching. Prior knowledge has been leveraged by modeling disease correlations using knowledge graphs, such as KG^[19], PPKED^[7], and RareGen^[10]. Some studies have compiled template databases to incorporate patterned information, such as Hybrid^[15] and R2Gen^[11]. Others have addressed data bias issues through contrastive learning mechanisms^[6] or curriculum learning^[4]. Several studies endeavors have sought to incorporate background data, including diagnostic details like medical histories and patient-reported symptoms, to facilitate the generation of medical reports^[41–43]. Furthermore, other studies have delved into addressing issues, such as semantic redundancy^[39], the coherence of report content^[18], and the integration of visual data from multiple perspectives^[13], among other challenges. Yin et al.^[39] proposed a topic-matching mechanism to mitigate semantic repetition issues. Xue et al.^[18] introduced a recurrent report generation model to ensure semantic coherence between adjacent sentences. Yan et al.^[21] developed a weakly supervised method that contrasts target reports with “hard” negative samples, i.e., incorrect reports.

2.3 RAG

RAG constitutes a methodology that fuses retrieval mechanisms with pre-trained models to amplify efficacy in generative tasks^[44]. RAG harnesses the linguistic comprehension and generative prowess of pre-trained models, while also integrating pertinent information retrieved to furnish precise and

comprehensive responses. This strategy was initially proposed by Ref. [45] and has demonstrated substantial advancements across numerous NLP challenges. Commonly, such models absorb knowledge by extracting relevant documents from an external memory using the input text as a query, with the language model then employing these documents to enhance its predictive capabilities^[46–48]. Recently, research efforts have explored the integration of multi-modal data into specialized tasks via RAG, including multi-modal language modeling^[49], visual question answering^[50], and contrastive language-image pre-training^[51], among others. Although there is a modest number of studies intersecting RAG with report generation, they do not fundamentally rely on RAG. For example, Li et al.^[15] introduced a hybrid report generation model that retrieves template sentences from a pre-existing template database concurrent with generating standard sentences. KERP^[17] constructs reports by rephrasing the retrieved text templates.

3 Method

3.1 Problem formulation

Before introducing our proposed method, we first establish the problem formulation. This paper utilizes an existing paired image-report dataset, denoted as $\mathcal{D}^r = \{(I_i^r, R_i)\}_{i=1}^M$, and a paired image-label dataset, $\mathcal{D}^l = \{(I_j^l, L_j)\}_{j=1}^N$. Here, I_i^r and R_i represent the i -th image and its corresponding report in the image-report dataset, respectively; I_j^l and L_j denote the j -th image and its corresponding semantic labels in the image-report dataset, respectively, with the semantic labels represented as multi-hot vectors; M and N represent the numbers of instances in $\mathcal{D}^r$ and $\mathcal{D}^l$, respectively. The objective is to train a report generation model, $f: (I_i^r, T_i^r) \mapsto R_i$, which maps a image I_i^r , and its corresponding retrieved semantic text T_i^r , to a textual medical report R_i , employing both datasets $\mathcal{D}^r$ and $\mathcal{D}^l$. Figure 2 shows the diagram of RAGSemRad.

Upon receiving the visual features of an input image I_i^r and its corresponding semantic text T_i^r retrieved by our proposed semantic retrieval module, they are embedded into the visual hidden matrix $H_{v,i}^r$ and the text hidden matrix $H_{s,i}^r$, using the standard Transformer encoder. Subsequently, an LLM that acts as the decoder processes $H_{v,i}^r$, $H_{s,i}^r$, and $\{\hat{y}_m\}_{m<t}$ to generate the token at time step t , which alleviates the data bias

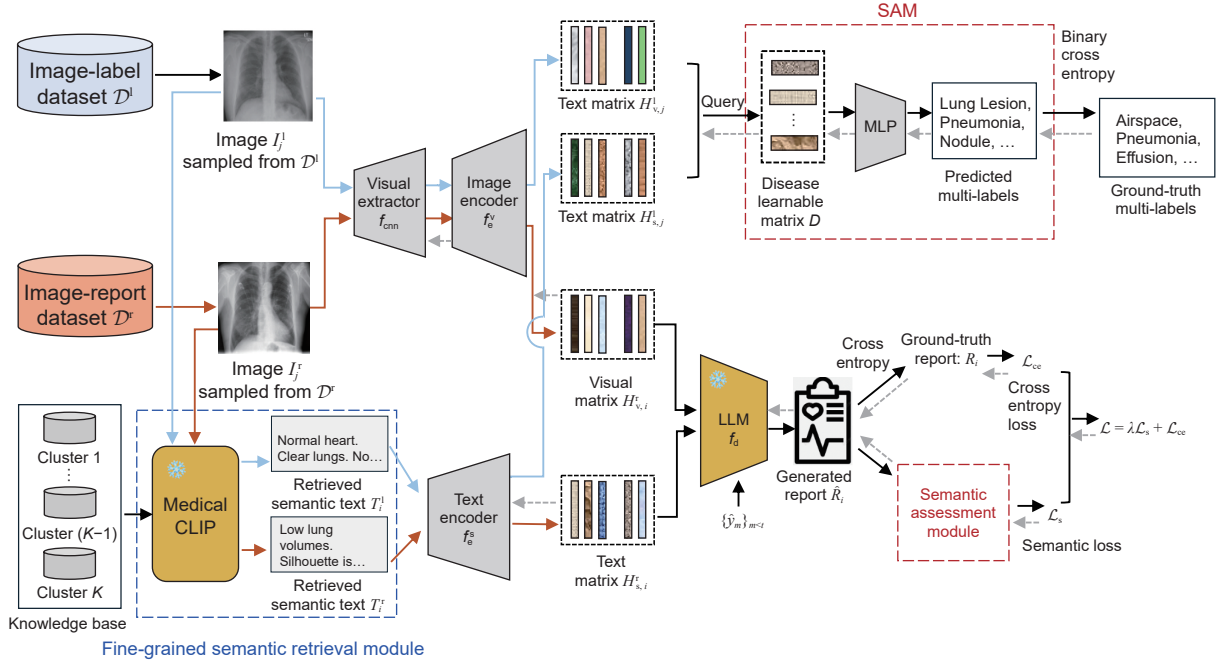


Fig. 2 Diagram of RAGSemRad. RAGSemRad is comprised of two key modules: the fine-grained semantic retrieval module, highlighted as the blue section, and the semantic assessment module, highlighted as the red section. The flow of gradients during back-propagation is shown with dotted light gray lines.

by leveraging the complementary semantic information contained in T_i^r . This process is formulated as follows:

$$H_{v,i}^r = f_e^v(V_i^r) \quad (1)$$

$$H_{s,i}^r = f_e^s(T_i^r) \quad (2)$$

$$\hat{y}_t = f_d(\{\hat{y}_m\}_{m < t}, H_{v,i}^r, H_{s,i}^r) \quad (3)$$

where $V_i^r \in \mathbf{R}^{c \times (h \times w)}$ represents the visual features of the input image I_i^r extracted by the visual extractor f_{cnn} , c , h , and w denote the number of channels, height, and width of V_i^r , respectively, obtained by tokenizing T_i^r and multiplying by a parameter matrix; f_e^v and f_e^s represent the standard encoders in the Transformer, each with distinct parameters; f_d denotes the decoder, an LLM; and $\hat{y}_t$ is the token generated at time step t . The generated report for the input image I_i^r is denoted as $\hat{R}_i = \{\hat{y}_1, \hat{y}_2, \dots, \hat{y}_{|\hat{R}_i|}\}$, where $|\hat{R}_i|$ signifies the length of $\hat{R}_i$.

To enhance the upper bound of the model's performance, we introduce the semantic assessment module, which utilizes supervision signals from the image-label data $\mathcal{D}^l$ to guide the training process. Essentially, it takes the generated report $\hat{R}_i$ as input and further enhances the performance by assessing the semantic correctness of $\hat{R}_i$.

In summary, our report generation model is

optimized by both the semantic loss derived from the semantic assessment module and the cross-entropy loss,

$$\mathcal{L} = \lambda \cdot \mathcal{L}_s + \mathcal{L}_{ce} \quad (4)$$

where λ is a hyperparameter that weights the two losses, $\mathcal{L}_s$ and $\mathcal{L}_{ce}$ represent the semantic loss and the cross-entropy loss, respectively.

In the following sections, we will provide detailed insights into the two key modules.

3.2 Fine-grained semantic retrieval module

3.2.1 Sentence-level clustering

We initiate sentence-level clustering to group sentences with similar or identical semantic information together. To begin, we shuffle the reports in the training dataset at the sentence level, considering that a medical report typically consists of multiple sentences, each addressing a specific medical observation^[15]. We start by generating embeddings for each sentence using the text encoder from Medical Contrastive Language-Image Pre-training (MedCLIP)^[52], which will be elaborated in the following subsection. Subsequently, we apply K -means clustering to categorize these sentences into K clusters. Following clustering, each cluster comprises sentences sharing similar or identical semantic information.

3.2.2 Semantic text retrieval

For retrieving the semantic text T_i , we employ a pre-trained vision-language model, MedCLIP^[52], which is akin to Contrastive Language-Image Pre-training (CLIP)^[53], but trained with a contrastive approach on medical image-text pairs. Notably, MedCLIP replaces CLIP's InfoNCE loss with a semantic matching loss, utilizing medical semantic similarity between each image and report as the supervision signal. Consequently, MedCLIP can identify a sentence with akin or identical semantic information for a given image. In particular, we curate a set of candidate sentences $\{S_{i,n}\}_{n=1}^K$ for each input image by selecting one sentence randomly from each cluster. The sentences in $\{S_{i,n}\}_{n=1}^K$ with the top- Q similarities to an image are considered the image's semantic sentences, thereby retrieving the image's semantic information. Subsequently, these retrieved sentences are concatenated to consist of the semantic text T_i . This process is depicted as follows:

$$\{S_{i,n}\}_{n=1}^Q = \text{top-}Q \langle f_v^c(I_i), f_t^c(S_{i,n}) \rangle_{0 \leq n \leq K} \quad (5)$$

$$T_i = \text{Concat}(\{S_{i,n}\}_{n=1}^Q) \quad (6)$$

where $f_v^c(\cdot)$ and $f_t^c(\cdot)$ denote the image encoder and text encoder of MedCLIP, respectively. It is crucial to note that the parameters of MedCLIP are fixed during this retrieval process. $\text{Concat}(\cdot)$ represents concatenation. It should be noted that the retrieved sentences are not simply concatenated in a straightforward manner. Instead, we leverage a large language model, General Language Model (GLM) utilized in our study, to extract and distill the most pertinent semantic fragments from the retrieved sentences. For example, the original sentence, "Persistent small right pleural effusion and XXXX XXXX atelectasis", is meticulously refined down to the essence "small right pleural effusion". This process effectively filters out any superfluous or noisy details, ensuring that the information retained is both concise and relevant to the task at hand.

3.3 Semantic assessment module

The Semantic Assessment Module (SAM) is designed to enhance the upper bound of RAGSemRad's performance by enhancing the alignment between the input image and the resulting output report, leveraging the supervision signals from paired image-label data.

Specifically, during the encoding phase, SAM

operates as a classifier, i.e., a learnable query matrix D with Multi-Layer Perceptron (MLP). It takes both the image I_j^1 from $\mathcal{D}^1$ and the corresponding retrieved semantic text T_j^1 as input, generating a probability distribution over the semantic labels, in which supervision signals are introduced. While assessing semantic correctness, SAM queries over the generated report $\hat{R}_i$ and produces a semantic loss indicating its semantic correctness. This loss is then backpropagated, so as to improve the upper bound of RAGSemRad's performance. To maintain differentiability and enable the flow of supervision signals, a soft approach is employed to derive E_i , i.e., the representation matrix of $\hat{R}_i$. This process is formulated as follows:

$$E_i = P_i W \quad (7)$$

$$\mathcal{L}_s = \text{MLP}(\text{Att}(D, E_i), E_i) \quad (8)$$

where $W \in \mathbf{R}^{|\mathcal{V}| \times e}$ denotes the vocabulary embedding matrix, with $|\mathcal{V}|$ and e representing the vocabulary size and word embedding dimension, respectively; and $P_i \in \mathbf{R}^{|\hat{R}_i| \times |\mathcal{V}|}$ is a matrix representing probabilities of $\hat{R}_i$ over the vocabulary words. SAM queries the generated report $\hat{R}_i$ by applying soft attention over E_i , resulting the semantic loss $\mathcal{L}_s$. In details, $\text{Att}(\cdot)$ represents the multi-head attention^[54] to model the query process; MLP, i.e., the classifier, is to produce the semantic loss. Therefore, SAM is the query matrix D combined with MLP.

It is crucial to emphasize that SAM is not utilized during the inference phase. Also, the parameters of the module remain fixed when evaluating the generated report $\hat{R}_i$. The training of RAGSemRad operates in an adversarial manner, where the report generation module serves as a generator producing reports, while the semantic assessment module functions as a discriminator to discriminate the semantic correctness of the generated reports $\hat{R}_i$. More details, the SAM is initially trained on the classification task to establish its capability in assessing the semantic correctness. The report generation module and the semantic assessment module engage in a process of alternating parameters fixing and updating. To put it more plainly, as the SAM enhances its ability to evaluate the semantic correctness, the report generation module, in turn, experiences an improvement in the semantic correctness of its generated reports. This iterative refinement process allows both components to work in

tandem, each one's development bolstering the performance of the other.

4 Experiment and Result

4.1 Datasets

For our experiments, we utilize both paired image-report datasets and a paired image-label dataset.

Image-label dataset: CheXpert. The image-label dataset we employed is CheXpert^[55], a publicly available dataset containing 224 316 radiology X-ray images along with 14-dimensional clinical labels, which include annotations for conditions: Atelectasis, Cardiomegaly, Consolidation, Edema, Enlarged Cardiomediastinum, Fracture, Lung Lesion, Lung Opacity, No Finding, Pleural Effusion, Pleural Other, Pneumonia, Pneumothorax, and Support Devices.

Image-report datasets: IU X-Ray and MIMIC-CXR. We utilize two benchmark datasets for report generation: IU X-Ray^[56] and MIMIC-CXR^[57]. IU X-Ray comprises 3955 reports and 7470 chest X-ray images from frontal and/or lateral views. Each study typically includes sections like impression, findings, comparison, and indication. For consistency with prior research^[1, 7, 11], we concatenate the impression and findings sections as the target output and utilize multi-view chest X-ray images. MIMIC-CXR, the largest report generation dataset to date, comprises 377 110 chest X-ray images and 227 835 reports from 64 588 patients. The dataset includes images from various views, such as Anterior-Posterior (AP), Posterior-Anterior (PA), and Lateral (LA). Reports in MIMIC-CXR usually consist of multiple sections, like comparison, clinical history, indication, reasons for examination, impressions, and findings. Following the experimental setups in Ref. [9], we focus on generating text from the “findings” section as the medical report and utilize multi-view images.

Consistent with previous studies^[1, 7, 11], we split the image-report pairs randomly for training, validation, and testing sets of IU X-Ray with proportions of 0.8, 0.1, and 0.1, respectively. For MIMIC-CXR, we adopt the official split: 368 960 in the training set, 2991 in the validation set, and 5159 in the testing set. The image-label dataset is included in the training sets of IU X-Ray and MIMIC-CXR. All evaluations are conducted on the testing sets of IU X-Ray and MIMIC-CXR, separately.

4.2 Metrics

Consistent with current research trends, we assess the efficacy by utilizing standard Natural Language Generation (NLG) evaluation metrics. These encompass BLEU-1, 2, 3, 4, and ROUGE-L scores, which are broadly used to gauge the quality of text. Furthermore, our evaluation scope is broadened to incorporate a report classification model, specifically for determining the Clinical Effectiveness (CE) of the reports produced. The metrics for evaluating the classification model consist of Accuracy, F1-score, Precision, and Recall. In order to enhance the reliability of our findings and reduce the influence of limited sample sizes on individual class metrics, we utilize the micro-average method to compute the Precision, Recall, and F1-score metrics.

4.3 Model settings

In our model setup, we utilize DenseNet 121^[58] as the visual extractor f_{cnn} for extracting visual features from input images, with the fully connected layers removed. For the image and text encoders, Multi-Head Attention (MHA) layer is employed, with the number of heads in MHA set to 3. To handle multi-label classification, we implement a weighted binary cross-entropy loss, transforming it into multiple binary classifications. We take the 3.55×10^8 -parameter model GPT-2 Medium^[59] as the decoder in this paper. Parameter optimization is carried out using the Adam optimizer with a batch size of 32 and a learning rate of 0.0001. Considering the average sentence count in a report, we select the top-7 sentences as the semantic text for each image. All implementations are executed using PyTorch and run on NVIDIA Tesla V100 GPUs. For fast retrieval, we utilize the FAISS package^[60], and the retrieval process is implemented as a pre-processing step. This ensures instantaneous retrieval during both training and inference, with negligible computational overhead. λ , the weight of the semantic loss term, is determined through grid search on the validation dataset, and it is set to be 0.3.

4.4 Baselines

We have carefully chosen a selection of representative and newly proposed methods for report generation as our baselines. These baselines cover a spectrum of strategies and innovations within the field, including Co-Att^[1], R2Gen^[11], KG^[19], CMM^[61], and MS-

Gen^[22], each offering distinct contributions and strengths. Co-Att^[1] is a pioneering multi-task learning architecture that has significantly propelled the progress of automatic report generation. R2Gen^[11] utilizes a memory module specifically crafted to retain key information throughout the report creation process. KG^[19] adopts an innovative strategy by building a knowledge graph in collaboration with domain specialists, which is then integrated into the model to deepen its understanding of medical terminology and their interconnections, crucial for the production of precise and contextually rich radiology reports. CMM^[61] introduces an original cross-modal memory framework designed to refine the synchronization of visual and textual attributes, thereby enhancing the model's capability to generate reports that are both visually informed and descriptively detailed. MS-Gen^[22] has achieved top-tier performance in the domain, incorporating sophisticated semantic analysis that enables it to generate reports not only accurate but also abundant in clinical insights, setting a challenging standard for comparison in our experimental assessments. Note that, considering that MS-Gen is the state-of-the-art method for current report generation and its architectural design shares similarities with our method RAGSemRad, we have chosen it as the primary comparison baseline instead of RAG-based report generation methods.

4.5 Result analysis

Table 1 presents a comprehensive overview of the quantitative experimental findings concerning RAGSemRad and its baseline models, focusing on a range of NLG metrics. Notably, RAGSemRad demonstrates commendable performance on both the IU X-Ray and MIMIC-CXR datasets, underscoring its effectiveness in the realm of report generation. This success can be primarily attributed to the adept integration of the proposed semantic retrieval module and semantic assessment module within RAGSemRad.

An interesting observation is that RAGSemRad surpasses the state-of-the-art MS-Gen method almost all language metrics when evaluated on both IU X-Ray and MIMIC-CXR, showcasing its superiority in generating reports. We can find RAGSemRad achieves slightly lower BLEU-1 and BLEU-4 on IU X-Ray and MIMIC-CXR, comparing with MS-Gen, with 0.496 vs. 0.494 on IU X-Ray and 0.124 vs. 0.118 on MIMIC-CXR. However, this does not imply that the performance of RAGSemRad is poor. It is crucial to emphasize that these metrics, namely BLEU-N and ROUGE-L, primarily focus on assessing word overlap between the generated reports and the reference reports, rather than capturing the semantic correctness of the generated reports. Furthermore, the presence of data biases poses a challenge, as a report generation model may achieve inflated BLEU-N scores by merely reproducing the most common sentences or terms

Table 1 Performance comparison of report generation on IU X-Ray and MIMIC-CXR. The second highest scores are underlined, and the black bold font indicates the best result obtained for the corresponding indicator.

Dataset	Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L
IU X-Ray	Co-Att ^[1]	0.450	0.278	0.189	0.136	0.355
	KG ^[19]	0.446	0.300	0.211	0.152	0.359
	R2Gen ^[11]	0.474	0.312	0.221	0.168	0.375
	CMM ^[61]	0.492	0.323	0.231	0.177	0.371
	MS-Gen ^[22]	0.496	0.330	<u>0.237</u>	<u>0.183</u>	<u>0.382</u>
	RAGSemRad (ours)	<u>0.494</u>	0.343	0.251	0.209	0.411
	Ours without RA	0.465	0.281	0.211	0.150	0.358
	Ours without SAM	0.480	0.322	0.225	0.179	0.373
MIMIC-CXR	Co-Att ^[1]	0.331	0.220	0.147	0.117	0.276
	KG ^[19]	0.341	0.221	0.136	0.119	0.272
	R2Gen ^[11]	0.356	0.223	0.151	0.110	0.283
	CMM ^[61]	<u>0.370</u>	0.228	<u>0.157</u>	0.107	0.285
	MS-Gen ^[22]	0.369	<u>0.231</u>	<u>0.157</u>	0.124	<u>0.295</u>
	RAGSemRad (ours)	0.372	0.242	0.166	<u>0.118</u>	0.303
	Ours without SR	0.336	0.191	0.134	0.103	0.250
	Ours without SAM	0.355	0.225	0.149	0.107	0.266

present in the training data. The evaluation of report classification results, as delineated in Table 2, serves as the validation of the superior clinical accuracy exhibited by the reports generated through RAGSemRad. This reinforces the notion that RAGSemRad not only excels in generating reports that adhere closely to the ground truth, but also produces outputs that are clinically accurate and relevant. It is observed that RAGSemRad outperforms MS-Gen by more than 3% and nearly 2% in terms of accuracy, showcasing its effectiveness in the realm of medical report generation. Additionally, RAGSemRad attains a superior Recall score on both the IU X-Ray and MIMIC-CXR datasets, confirming its strength in mitigating the issue of data bias.

4.6 Ablation study

The ablation study results are summarized in Table 1 for report generation and Table 2 for report classification. Across both NLG and classification metrics, RAGSemRad consistently outperforms the variants without the semantic retrieval (w/o SR) and semantic assessment module (w/o SAM) modules, underscoring the efficacy of integrating these components into the model architecture. Notably, when comparing the complete RAGSemRad model with the variant lacking the semantic assessment module (w/o SAM), substantial performance improvements are observed. For instance, on the IU X-Ray dataset, the

ROUGE-L score increases from 0.373 to 0.411, and on MIMIC-CXR, it rises from 0.266 to 0.303, highlighting the benefits of leveraging supervision signals from paired image-label data.

The results depicted in Table 2 further validate the significance of utilizing image-label data in enhancing performance across all classification evaluation metrics. This enhancement signifies that the reports generated by RAGSemRad are more accurate in terms of semantic content. When comparing RAGSemRad with the variant lacking the semantic retrieval module (w/o SR), improvements in both NLG and classification metrics are evident, emphasizing the advantages of the semantic retrieval module in mitigating data biases and enhancing overall performance. For instance, the accuracy increases from 0.357 to 0.392, and the recall increases from 0.348 to 0.397.

Moreover, to delve deeper into the performance of RAGSemRad, we conducted a qualitative examination, and the findings are detailed in Fig. 3. It is evident that in the three presented cases, RAGSemRad stands out as the top performer, accurately identifying all anomalies. This is attributed to the retrieval-augmented module, which furnishes RAGSemRad with precise semantic information, while the semantic supervision module guarantees the alignment of these semantics. KG, on the other hand, fares the worst, failing to detect even a single anomaly, which points to the constraints of a

Table 2 Performance comparison of report classification on IU X-Ray and MIMIC-CXR. The black bold font indicates the best result obtained for the corresponding indicator.

Dataset	Method	Accuracy	F1-score	Precision	Recall
IU X-Ray	Co-Att ^[1]	0.365	0.386	0.451	0.351
	KG ^[19]	0.350	0.370	0.422	0.330
	R2Gen ^[11]	0.356	0.377	0.426	0.332
	CMM ^[61]	0.363	0.408	0.465	0.351
	MS-Gen ^[22]	0.376	0.418	0.478	0.359
	RAGSemRad (ours)	0.392	0.451	0.502	0.397
	Ours w/o SR	0.357	0.400	0.452	0.348
	Ours w/o SAM	0.371	0.411	0.470	0.352
MIMIC-CXR	Co-Att ^[1]	0.371	0.403	0.456	0.351
	KG ^[19]	0.385	0.413	0.458	0.369
	R2Gen ^[11]	0.373	0.416	0.454	0.378
	CMM ^[61]	0.371	0.419	0.453	0.382
	MS-Gen ^[22]	0.412	0.435	0.470	0.400
	RAGSemRad (ours)	0.443	0.477	0.511	0.431
	Ours w/o SR	0.377	0.415	0.449	0.386
	Ours w/o SAM	0.393	0.436	0.478	0.398

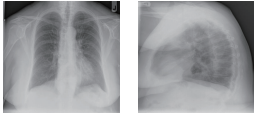
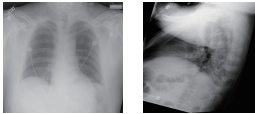
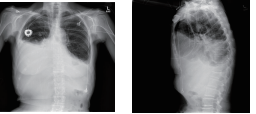
Method			
Ground truth	The heart size and pulmonary vascularity appear within normal limits. A large hiatal hernia is noted. The lungs are free of focal airspace disease. No pneumothorax or pleural effusion is seen. Degenerative changes are present in the spine.	The cardiomeastinal silhouette is normal in size and contour. There are a few XXXX opacities in the lung bases bilaterally. No definitive pneumothorax or pleural effusion. Displaced fracture of the mid one-third of the right clavicle.	There is a moderate sized right pleural effusion. Severe slightly smaller than is compared to XXXX. There is a small left pleural effusion. This is unchanged as compared to the prior study. There is a right chest wall venous XXXX XXXX which appears accessed. No pneumothorax. Scaphoid abdomen.
KG	The lungs are clear. There is no pleural effusion. The heart and mediastinum are normal as are the skeletal structures and soft tissues.	The heart size and pulmonary vascularity appear within normal limits. Lungs are free of focal airspace disease. No pleural effusion or pneumothorax is seen.	The heart size is normal. The lungs are free of any focal infiltrates. There are no nodules or masses. There is no pneumothorax. There is no pleural effusion.
MS-Gen	The heart size and pulmonary vascularity appear within normal limits. Lungs are free of focal airspace disease. Hiatal hernia is noted. No pleural effusion or pneumothorax is seen. Calcified granuloma are noted.	Heart size and mediastinal contours appear within normal limits. No focal lung consolidation, pleural effusion or pneumothorax. No acute bony abnormality.	The thoracic aorta is calcified. No focal areas of pulmonary consolidation. There is pleural effusion. No pneumothorax. No pleural effusion. The thoracic spine appears intact.
RAGSemRad	Heart size within normal limits. Mild hyperinflation appears similar to prior. No focal alveolar consolidation. Definite hiatal hernia is seen.	Lungs are without focal consolidation, effusion or pneumothorax. Normal heart size. Opacities is identified. No typical findings of pulmonary edema.	Heart size is normal. There is pneumothorax. Evident of pleural effusion. No pneumothorax. No acute bony abnormality.
Retrieved Semantic information	Hiatal hernia Normal heart size No consolidation Without focal consolidation No pleural effusion No pneumothorax Lung is normal	Normal heart size No consolidation Opacities in lung No pneumothorax Clear lung Normal cardiomeastinal No focal volume	No focal consolidation Clear lungs No acute bony abnormality Intact bony structures Pleural effusion Normal mediastinum Normal osseous structures

Fig. 3 Three examples of ground-truth reports alongside the reports generated by RAGSemRad (our method) and baselines. The sentences highlighted in blue indicate the diseases present in the respective cases. The terms in red denote the diseases accurately identified in the generated reports.

knowledge graph with limited coverage of disease nodes. In the second case, it is noteworthy that even the leading method MS-Gen fails to identify the anomaly, likely due to the semantics not being adequately captured. Furthermore, the alignment between the semantics retrieved and the final report produced highlights RAGSemRad's advantage in terms of interpretability.

4.7 Impact of sampling rate

As the results shown in Fig. 4, we can observe that as the sampling rate increases, the performance of the model shows a clear upward trend, and the possible reasons for the improvement in model performance mainly include the following aspects: Firstly, more accurate supervision signals enable the model to acquire more corresponding relationships between images and reports. These signals provide refined guidance for the model, enhancing its understanding of the semantic associations between image content and reports. Secondly, the enhanced semantic understanding, through the SAM utilizing additional supervision signals, effectively trains the model to recognize and distinguish key features, improving the

model's comprehension of image content. Meanwhile, more image-label data help to reduce the model's dependence on specific samples, decrease the risk of overfitting, and ensure that the model maintains good performance on unseen data. Additionally, the increase in data diversity improves the model's generalization ability, allowing it to learn more generalized features and adapt to different scenarios. Optimizing the loss function during the training process is also key to improving performance, as the additional supervision information promotes rapid convergence of the model. Finally, improving the anomaly detection capability is crucial for report generation tasks. The abundant image-label data enable the model to more accurately identify and handle abnormal situations, generating more precise reports.

5 Limitations and Future Work

RAGSemRad stands out as a report generation model that leverages RAG, where the retrieval step plays a pivotal role in determining its overall performance. The effectiveness of RAGSemRad hinges on its ability to retrieve information that is not only relevant but also crucial to the query at hand. In the context of

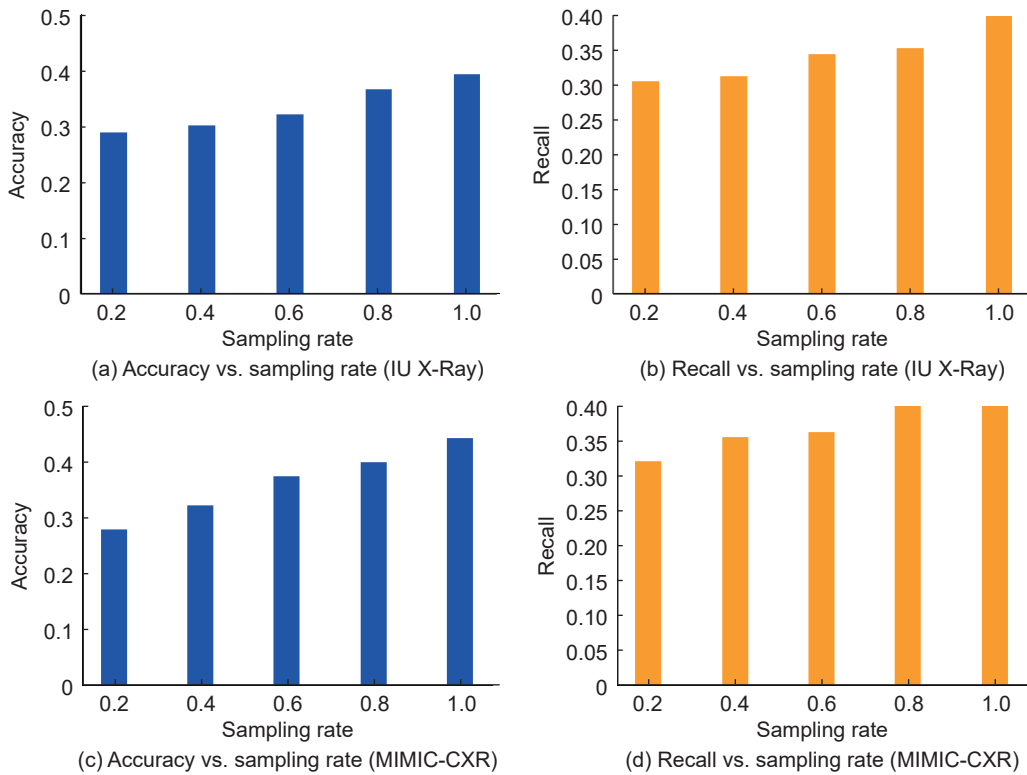


Fig. 4 Model's performance in terms of accuracy and recall metrics showing an upward trend for both the IU X-Ray and MIMIC-CXR datasets as the image-label sampling rate increases. A comprehensive analysis is presented Section 4.7.

clustering, the determination of the number of clusters often relies on empirical experience. It is crucial to acknowledge that during the semantic retrieval process, semantic noises might infiltrate the system, leading to potential inaccuracies that could misguide the model in generating erroneous or imprecise sentences. Addressing these challenges and mitigating the impact of semantic noises will be critical focal points in refining the performance and reliability of RAGSemRad in future iterations.

6 Conclusion

In this research, we propose a pioneering radiology report generation model, namely RAGSemRad. This innovative model marks a significant step forward in the realm of automated radiology report generation by harnessing the complementary semantic information encapsulated within the dataset and synergizing it with the valuable insights derived from the image-label dataset. The incorporation of these dual sources of semantic information not only enriches the report generation process, but also opens up new avenues for more efficient and reliable automated radiology reporting systems.

Acknowledgment

This work was supported by the Fundamental Research Funds for the Central Universities (No. 2232025D-34), the Noncommunicable Chronic Diseases-National Science and Technology Major Project (Nos. 2024ZD0532400 and 2024ZD0532403), the Sichuan Provincial Science and Technology Program Key Research and Development Project (No. 2024YFFK0443), and the National Natural Science Foundation of China (Nos. 62477006 and 61975124).

References

- [1] B. Jing, P. Xie, and E. Xing, On the automatic generation of medical imaging reports, in *Proc. 56th Annu. Meeting of the Association for Computational Linguistics*, Melbourne, Australia, 2018, pp. 2577–2586.
- [2] O. Vinyals, A. Toshev, S. Bengio, and D. Erhan, Show and tell: A neural image caption generator, in *Proc. 2015 IEEE Conf. Computer Vision and Pattern Recognition*, Boston, MA, USA, 2015, pp. 3156–3164.
- [3] J. Krause, J. Johnson, R. Krishna, and F-F. Li, A hierarchical approach for generating descriptive image paragraphs, in *Proc. 2017 IEEE Conf. Computer Vision and Pattern Recognition*, Honolulu, HI, USA, 2017, pp. 3337–3345.

- [4] F. Liu, S. Ge, and X. Wu, Competence-based multimodal curriculum learning for medical report generation, in *Proc. 59th Annu. Meeting of the Association for Computational Linguistics and the 11th Int. Joint Conf. Natural Language Processing*, Virtual Event, 2021, pp. 3001–3012.
- [5] D. You, F. Liu, S. Ge, X. Xie, J. Zhang, and X. Wu, Aligntransformer: Hierarchical alignment of visual regions and disease tags for medical report generation, in *Proc. 24th Int. Conf. Medical Image Computing and Computer-Assisted Intervention*, Strasbourg, France, 2021, pp. 72–82.
- [6] F. Liu, C. Yin, X. Wu, S. Ge, P. Zhang, and X. Sun, Contrastive attention for automatic chest X-ray report generation, in *Proc. Findings of the Association for Computational Linguistics*, Virtual Event, 2021, pp. 269–280.
- [7] F. Liu, X. Wu, S. Ge, W. Fan, and Y. Zou, Exploring and distilling posterior and prior knowledge for radiology report generation, in *Proc. 2021 IEEE/CVF Conf. Computer Vision and Pattern Recognition*, Nashville, TN, USA, 2021, pp. 13748–13757.
- [8] F. Liu, C. You, X. Wu, S. Ge, S. Wang, and X. Sun, Auto-encoding knowledge graph for unsupervised medical report generation, in *Proc. 35th Int. Conf. Neural Information Processing Systems*, Virtual Event, 2021, p. 1244.
- [9] H. Nguyen, D. Nie, T. Badamdorj, Y. Liu, Y. Zhu, J. Truong, and C. Li, Automated generation of accurate & fluent medical X-ray reports, in *Proc. 2021 Conf. Empirical Methods in Natural Language Processing*, Virtual Event, 2021, pp. 3552–3569.
- [10] X. Jia, Y. Xiong, J. Zhang, Y. Zhang, and Y. Zhu, Few-shot radiology report generation for rare diseases, in *Proc. 2020 IEEE Int. Conf. Bioinformatics and Biomedicine*, Seoul, Republic of Korea, 2020, pp. 601–608.
- [11] Z. Chen, Y. Song, T. H. Chang, and X. Wan, Generating radiology reports via memory-driven transformer, in *Proc. 2020 Conf. Empirical Methods in Natural Language Processing*, Virtual Event, 2020, pp. 1439–1449.
- [12] P. Harzig, Y. Y. Chen, F. Chen, and R. Lienhart, Addressing data bias problems for chest X-ray image report generation, in *Proc. 30th British Machine Vision Conf.*, Cardiff, UK, 2019, p. 144.
- [13] J. Yuan, H. Liao, R. Luo, and J. Luo, Automatic radiology report generation based on multi-view image fusion and medical concept enrichment, in *Proc. 22nd Int. Conf. Medical Image Computing and Computer-Assisted Intervention*, Shenzhen, China, 2019, pp. 721–729.
- [14] G. Liu, T. M. H. Hsu, M. McDermott, W. Boag, W. H. Weng, P. Szolovits, and M. Ghassemi, Clinically accurate chest X-ray report generation, in *Proc. 4th Machine Learning for Healthcare Conf.*, Ann Arbor, MI, USA, 2019, pp. 249–269.
- [15] C. Y. Li, X. Liang, Z. Hu, and E. P. Xing, Hybrid retrieval-generation reinforced agent for medical image report generation, in *Proc. 32nd Int. Conf. Neural Information Processing Systems*, Montréal, Canada, 2018, pp. 1537–1547.
- [16] Y. Miura, Y. Zhang, E. B. Tsai, C. P. Langlotz, and D. Jurafsky, Improving factual completeness and consistency of image-to-text radiology report generation, in *Proc. 2021 Conf. North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Virtual Event, 2021, pp. 5288–5304.
- [17] C. Y. Li, X. Liang, Z. Hu, and E. P. Xing, Knowledge-driven encode, retrieve, paraphrase for medical image report generation, in *Proc. 33rd AAAI Conf. Artificial Intelligence*, Honolulu, HI, USA, 2019, pp. 6666–6673.
- [18] Y. Xue, T. Xu, L. R. Long, Z. Xue, S. Antani, G. R. Thoma, and X. Huang, Multimodal recurrent model with attention for automated radiology report generation, in *Proc. 21st Int. Conf. Medical Image Computing and Computer-Assisted Intervention*, Granada, Spain, 2018, pp. 457–466.
- [19] Y. Zhang, X. Wang, Z. Xu, Q. Yu, A. Yuille, and D. Xu, When radiology report generation meets knowledge graph, in *Proc. 34th AAAI Conf. Artificial Intelligence*, New York, NY, USA, 2020, pp. 12910–12917.
- [20] D. Bahdanau, K. Cho, and Y. Bengio, Neural machine translation by jointly learning to align and translate, in *Proc. 3rd Int. Conf. Learning Representations*, San Diego, CA, USA, <https://dblp.org/db/conf/iclr/iclr2015.html#BahdanauCB14>, 2015.
- [21] A. Yan, Z. He, X. Lu, J. Du, E. Chang, A. Gentili, J. McAuley, and C. N. Hsu, Weakly supervised contrastive learning for chest X-ray report generation, in *Proc. Findings of the Association for Computational Linguistics*, Punta Cana, Dominican Republic, 2021, pp. 4009–4015.
- [22] X. Jia, Y. Xiong, Y. Zhang, and L. Li, Multi-modal fusion with semantic supervision for radiology report generation, in *Proc. 26th European Conf. Artificial Intelligence*, Kraków, Poland, 2023, pp. 1140–1147.
- [23] M. Li, B. Lin, Z. Chen, H. Lin, X. Liang, and X. Chang, Dynamic graph enhanced contrastive learning for chest X-ray report generation, in *Proc. 2023 IEEE/CVF Conf. Computer Vision and Pattern Recognition*, Vancouver, Canada, 2023, pp. 3334–3343.
- [24] J. Li, R. R. Selvaraju, A. D. Gotmare, S. Joty, C. Xiong, and S. C. H. Hoi, Align before fuse: Vision and language representation learning with momentum distillation, in *Proc. 35th Conf. Neural Information Processing Systems*, Virtual Event, pp. 9694–9705, 2021.
- [25] S. Song, A. Subramanyam, I. Madejski, and R. L. Grossman, LaB-RAG: Label boosted retrieval augmented generation for radiology report generation, arXiv preprint arXiv: 2411.16523, 2024.
- [26] S. Wang, Z. Zhao, X. Ouyang, Q. Wang, and D. Shen, ChatCAD: Interactive computer-aided diagnosis on medical image using large language models, arXiv preprint arXiv: 2302.07257, 2023.
- [27] M. Ranjit, G. Ganapathy, R. Manuel, and T. Ganu, Retrieval augmented chest X-ray report generation using OpenAI GPT models, in *Proc. 8th Machine Learning for Healthcare Conf.*, New York, NY, USA, 2023, pp. 650–666.
- [28] M. L. Bernardi and M. Cimitile, Report generation from

- X-ray imaging by retrieval-augmented generation and improved image-text matching, in *Proc. 2024 Int. Joint Conf. Neural Networks*, Yokohama, Japan, 2024, pp. 1–8.
- [29] J. Jeong, K. Tian, A. Li, S. Hartung, S. Adithan, F. Behzadi, J. Calle, D. Osayande, M. Pohlen, P. Rajpurkar, Multimodal image-text matching improves retrieval-based chest X-ray report generation, in *Proc. Medical Imaging with Deep Learning*, Nashville, TN, USA, 2023, pp. 978–990.
- [30] X. Yang, K. Tang, H. Zhang, and J. Cai, Auto-encoding scene graphs for image captioning, in *Proc. 2019 IEEE/CVF Conf. Computer Vision and Pattern Recognition*, Long Beach, CA, USA, 2019, pp. 10677–10686.
- [31] Z. J. Zha, D. Liu, H. Zhang, Y. Zhang, and F. Wu, Context-aware visual policy network for fine-grained image captioning, *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 2, pp. 710–722, 2022.
- [32] G. Li, L. Zhu, P. Liu, and Y. Yang, Entangled transformer for image captioning, in *Proc. 2019 IEEE/CVF Int. Conf. Computer Vision*, Seoul, Republic of Korea, 2019, pp. 8927–8936.
- [33] Q. You, H. Jin, Z. Wang, C. Fang, and J. Luo, Image captioning with semantic attention, in *Proc. 2016 IEEE Conf. Computer Vision and Pattern Recognition*, Las Vegas, NV, USA, 2016, pp. 4651–4659.
- [34] M. Cornia, M. Stefanini, L. Baraldi, and R. Cucchiara, Meshed-memory transformer for image captioning, in *Proc. 2020 IEEE/CVF Conf. Computer Vision and Pattern Recognition*, Seattle, WA, USA, 2020, pp. 10575–10584.
- [35] K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. S. Zemel, and Y. Bengio, Show, attend and tell: Neural image caption generation with visual attention, in *Proc. 32nd Int. Conf. Machine Learning*, Lille, France, 2015, pp. 2048–2057.
- [36] P. Zhu, X. Wang, L. Zhu, Z. Sun, W. S. Zheng, Y. Wang, and C. Chen, Prompt-based learning for unpaired image captioning, *IEEE Trans. Multim.*, vol. 26, pp. 379–393, 2024.
- [37] Y. Xie, L. Zhou, X. Dai, L. Yuan, N. Bach, C. Liu, and M. Zeng, Visual clues: Bridging vision and language foundations for image paragraph captioning, in *Proc. 36th Int. Conf. Neural Information Processing Systems*, New Orleans, LA, USA, 2022, p. 1257.
- [38] X. Xie, Y. Xiong, P. S. Yu, K. Li, S. Zhang, and Y. Zhu, Attention-based abnormal-aware fusion network for radiology report generation, in *Proc. 2019 Int. Conf. Database Systems for Advanced Applications*, Chiang Mai, Thailand, 2019, pp. 448–452.
- [39] C. Yin, B. Qian, J. Wei, X. Li, X. Zhang, Y. Li, and Q. Zheng, Automatic generation of medical imaging diagnostic report with hierarchical recurrent neural network, in *Proc. 2019 IEEE Int. Conf. Data Mining*, Beijing, China, 2019, pp. 728–737.
- [40] Z. Wang, L. Zhou, L. Wang, and X. Li, A self-boosting framework for automated radiographic report generation, in *Proc. 2021 IEEE/CVF Conf. Computer Vision and Pattern Recognition*, Nashville, TN, USA, 2021, pp. 2433–2442.
- [41] X. Wang, Y. Peng, L. Lu, Z. Lu, and R. M. Summers, TieNet: Text-image embedding network for common thorax disease classification and reporting in chest X-rays, in *Proc. 2018 IEEE/CVF Conf. Computer Vision and Pattern Recognition*, Salt Lake City, UT, USA, 2018, pp. 9049–9058.
- [42] Z. Zhang, P. Chen, M. Sapkota, and L. Yang, TandemNet: Distilling knowledge from medical images using diagnostic reports as optional semantic references, in *Proc. 20th Int. Conf. Medical Image Computing and Computer-Assisted Intervention*, Quebec City, Canada, 2017, pp. 320–328.
- [43] X. Huang, F. Yan, W. Xu, and M. Li, Multi-attention and incorporating background information model for chest X-ray image report generation, *IEEE Access*, vol. 7, pp. 154808–154817, 2019.
- [44] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, H. Wang, and H. Wang, Retrieval-augmented generation for large language models: A survey, arXiv preprint arXiv: 2312.10997, 2024.
- [45] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. T. Yih, T. Rocktäschel, et al., Retrieval-augmented generation for knowledge-intensive NLP tasks, in *Proc. 34th Int. Conf. Neural Information Processing Systems*, Vancouver, Canada, 2020, p. 793.
- [46] M. Yasunaga, J. Leskovec, and P. Liang, LinkBERT: Pretraining language models with document links, in *Proc. 60th Annu. Meeting of the Association for Computational Linguistics*, Dublin, Ireland, 2022, pp. 8003–8016.
- [47] S. Borgeaud, A. Mensch, J. Hoffmann, T. Cai, E. Rutherford, K. Millican, G. B. Van Den Driessche, J. B. Lespiau, B. Damoc, A. Clark, et al., Improving language models by retrieving from trillions of tokens, in *Proc. 39th Int. Conf. Machine Learning*, Baltimore, MD, USA, 2022, pp. 2206–2240.
- [48] T. Xie, C. H. Wu, P. Shi, R. Zhong, T. Scholak, M. Yasunaga, C. S. Wu, M. Zhong, P. Yin, S. I. Wang, et al., UnifiedSKG: Unifying and multi-tasking structured knowledge grounding with text-to-text language models, in *Proc. 2022 Conf. Empirical Methods in Natural Language Processing*, Abu Dhabi, United Arab Emirates, 2022, pp. 602–631.
- [49] M. Yasunaga, A. Aghajanyan, W. Shi, R. James, J. Leskovec, P. Liang, M. Lewis, L. Zettlemoyer, and W. T. Yih, Retrieval-augmented multimodal language modeling, in *Proc. 40th Int. Conf. Machine Learning*, Honolulu, HI, USA, 2022, pp. 39755–39769.
- [50] W. Lin, and B. Byrne, Retrieval augmented visual question answering with outside knowledge, in *Proc. 2022 Conf. Empirical Methods in Natural Language Processing*, Abu Dhabi, United Arab Emirates, 2022, pp. 11238–11254.
- [51] C. W. Xie, S. Sun, X. Xiong, Y. Zheng, D. Zhao, and J. Zhou, RA-CLIP: Retrieval augmented contrastive language-image pre-training, in *Proc. 2023 IEEE/CVF Conf. Computer Vision and Pattern Recognition*,

- Vancouver, Canada, 2023, pp. 19265–19274.
- [52] Z. Wang, Z. Wu, D. Agarwal, and J. Sun, MedCLIP: Contrastive learning from unpaired medical images and text, in *Proc. 2022 Conf. Empirical Methods in Natural Language Processing*, Abu Dhabi, United Arab Emirates, 2022, pp. 3876–3887.
- [53] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in *Proc. 38th Int. Conf. Machine Learning*, Virtual Event, 2021, pp. 8748–8763.
- [54] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, Attention is all you need, in *Proc. 31st Int. Conf. Neural Information Processing Systems*, Long Beach, CA, USA, 2017, pp. 6000–6010.
- [55] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghighi, R. Ball, K. Shpanskaya, et al., CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison, in *Proc. 33rd AAAI Conf. Artificial Intelligence*, Honolulu, HI, USA, 2019, pp. 590–597.
- [56] D. Demner-Fushman, M. D. Kohli, M. B. Rosenman, S. E. Shooshan, L. Rodriguez, S. Antani, G. R. Thoma, and C. J. McDonald, Preparing a collection of radiology examinations for distribution and retrieval, *J. Am. Med. Inf. Assoc.*, vol. 23, no. 2, pp. 304–310, 2016.
- [57] A. E. W. Johnson, T. J. Pollard, N. R. Greenbaum, M. P. Lungren, C. Y. Deng, Y. Peng, Z. Lu, R. G. Mark, S. J. Berkowitz, and S. Horng, MIMIC-CXR-JPG, a large publicly available database of labeled chest radiographs, arXiv preprint arXiv: 1901.07042, 2019.
- [58] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, Densely connected convolutional networks, in *Proc. 2017 IEEE Conf. Computer Vision and Pattern Recognition*, Honolulu, HI, USA, 2017, pp. 2261–2269.
- [59] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, Language models are unsupervised multitask learners, <https://huggingface.co/openai-community/gpt2-medium>, 2025.
- [60] J. Johnson, M. Douze, and H. Jégou, Billion-scale similarity search with GPUs, *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [61] H. Qin and Y. Song, Reinforced cross-modal alignment for radiology report generation, in *Proc. Findings of the Association for Computational Linguistics*, Dublin, Ireland, 2022, pp. 448–458.



Xing Jia received the PhD degree from Fudan University, China in 2024. He is currently a master supervisor at School of Computer Science and Technology, Donghua University, China. His research interests include image analysis, few-shot learning, and multi-modal learning.



Songwen Pei received the PhD degree in computer and software theory from Fudan University, China in 2009. He is currently a professor at Faculty of Intelligence Technology, Shanghai Institute of Technology, Shanghai, China. His main research interest is medical image analysis.



Cairong Yan received the PhD degree from Xi'an Jiaotong University, China in 2006. She is currently a professor and master supervisor at Donghua University, China. Her research interests include data mining, machine learning, parallel computing, distributed system, recommendation systems, information retrieval and mining, and smart education.



Yun Xiong received the PhD degree in computer and software theory from Fudan University, China in 2008. She is currently a professor of computer science at Fudan University, China. Her research interests include image analysis and data mining.



Yumeng Zhang received the PhD degree in international tourism management from City University of Macau, China in 2023. She is currently a full-time faculty member at Guangdong University of Science and Technology, China. Her research interests focus on the field of management, with the aim of exploring innovative theories and practical applications in this discipline.



Zhijun Fang received the PhD degree from Shanghai Jiao Tong University, China. He is currently a professor at School of Computer Science and Technology, Donghua University, China. His research interests include machine vision and intelligent data analysis.